6.9	Further planetary warming risks large-scale release of stored carbon, requiring responses from different players at different scales.	a.	Future emissions, atmospheric concentration levels and climate warming are uncertain owing to natural factors (the role of carbon sinks), human factors (economic growth, population, energy sources) and positive feedback mechanisms (carbon release from peatlands and permafrost, and tipping points, including forest die back and alterations to the thermohaline circulation). (8) ***(F: uncertainty of global projections)***
		b.	Adaptation strategies for a changed climate (water conservation and management, resilient agricultural systems, land-use planning, flood-risk management) have different costs and risks.
		c.	Re-balancing the carbon cycle could be achieved through mitigation (carbon taxation, renewable switching, energy efficiency, afforestation, carbon capture and storage, and solar radiation management) but this requires global scale agreement and national actions both of which have proved to be problematic. ***(A: attitudes of different countries, TNCs and people)***

Guidance for integrating geographical skills for Topic 6

Quantitative geographical skills must be developed in this topic, including understanding of simple mass balance, unit conversions and the analysis and presentation of data. Qualitative approaches may be used if appropriate. The following guidance on integrating geographical skills gives suggestions for opportunities to meet these requirements.

These skills are **not** exclusive to the topic areas under which they appear; students will need to be able to apply these skills across any suitable topic area throughout their course of study. *Appendix 1* outlines the full range of geographical skills.

(1) Use of proportional flow diagrams showing carbon fluxes.

(2) Use of maps showing global temperature and precipitation distribution.

(3) Graphical analysis of the energy mix of different countries, including change over time.

(4) Analysis of maps showing global energy trade and flows.

(5) Comparisons of emissions from different energy source.

(6) Using GIS to map land-use changes such as deforestation over time.

(7) Analysis of climate model maps to identify areas at most risk from water shortages, floods in the future.

(8) Plotting graphs of carbon dioxide levels, calculating means and rates of change.

Area of study 4: Human Systems and Geopolitics